Annex I. Debt Sustainability Analysis

Risk of external debt distress:	Moderate
Augmented by significant risks stemming from domestic public and/or private external debt?	No

The debt sustainability analysis (DSA) shows improved prospects relative to the previous update.[1] The main factors for this assessment are the strong fiscal consolidation—containing high-quality fiscal measures—and improved external conditions. However, new uncertainties arose due to the designation by the U.S. Treasury's Office of Foreign Assets Control (OFAC) pursuant to the Kingpin act of some companies within the Grupo Continental (GC) and its owners. Staff's assessment is that under the current circumstances, Honduras' risk of external debt distress is assessed to remain moderate. External public debt indicators, except for the most extreme shock in only one indicator, are expected to remain below their indicative thresholds.[2] Against this background, an improved rating of low risk of debt distress could be considered in a future DSA provided that the program's fiscal consolidation and growth targets are achieved.

A. Background

1. Honduras' public debt increased slightly in 2014. After strong implementation of the program, the total public debt stood at 46.4 percent of GDP in 2014, a small increase of ¾ percentage points of GDP relative to 2013. The aggressive reduction in the public deficit in 2014 *vis-à-vis* 2013 (from 7.6 to 4.3 percent of GDP) explained this result.

2. The measure of public debt reported in Honduras is comprehensive. Public debt is defined as the gross stock of liabilities from the non-financial public sector (general government and public companies) owned by private investors plus the stock of central bank securities held by private banks (around 5½ percent of GDP).[3] The general government's debt includes arrears (2½ percent of GDP), and PPPs (0.1 percent of GDP). Since 2015, the fiscal reporting of PPPs is based on the IPSAS-32 standard.[4] Under this standard, PPP operations are reported in the fiscal accounts based on ownership criteria, and not on financing.

[1] See *Appendix I* of the *Staff Report for a Request for a Stand-by Arrangement and an Arrangement under the Standby Credit Facility* (IMF Country Report No. 14/361).

[2] Honduras three-year average CPIA score is 3.41 and its policy performance category is medium.

[3] These are securities issued for liquidity regulation and recorded as liabilities in the central bank's balance sheet.

[4] The stock of arrears is estimated. The authorities are currently implementing an audit with the help of an international audit firm to identify valid claims.

3. Public debt is mostly with foreign creditors. About 60 percent of total public debt is external but the share of domestic debt has risen in recent years. The main external lenders to Honduras are the Inter-American Development Bank (IADB), the Central American Bank for Economic Integration, and the World Bank. Debt to these institutions carries long maturities but only the IADB continues to provide loans on concessional terms (i.e., with a grant element of at least 35 percent). Domestic public debt is mainly to commercial banks, has a shorter—though rising—maturity (about 3 years), and carries a higher real interest rate (Tables A1 and A2). The authorities are implementing a debt management operation, seeking to refinanced US$530 million. Currently 78 percent has been refinanced. The recent creation of a private pension fund scheme is expected to increase the demand for domestic securities thus deepening and expanding the domestic capital market.

B. Underlying Assumptions

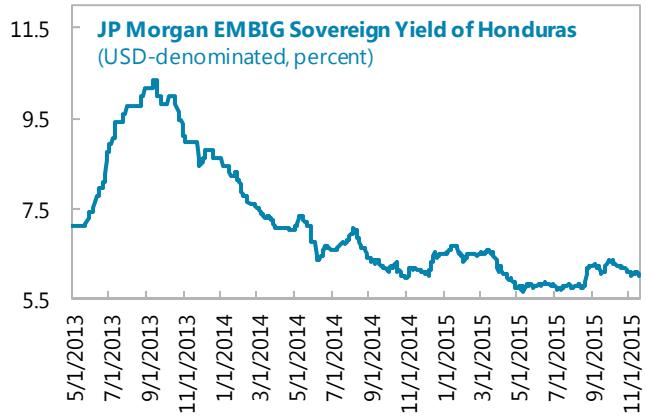
4. Economic growth. The baseline scenario assumes that growth in Honduras will rise to 3.5 percent in 2015 and 3.5 percent in 2016 and strengthen further in subsequent years, before stabilizing at about 4 percent—broadly in line with its current estimated potential. This reflects an improvement from the previous DSA, where growth was expected to reach 3¾ percent in the medium term. The acceleration of growth over the medium term is predicated on more favorable external conditions than previously expected (notably better terms of trade, lower fuel prices and a growth pick up in the U.S.), the recovery of the coffee sector from leaf-rust disease, as well as improved confidence and higher private investment from stronger economic policy management. These factors are expected to offset the dampening effect of fiscal consolidation on growth.

5. Fiscal policy. The baseline scenario incorporates a cumulative fiscal adjustment (measured using the primary balance) of about 6¾ percent of GDP during the program period (2014–17), with the primary deficit of the CPS declining to ½ percent of GDP by 2017. The overall CPS deficit continues to decline slowly over the medium term as debt and interest payments start to fall, stabilizing at about 1 percent of GDP over the longer term (consistent with a small primary surplus).[5] The fiscal adjustment features a combination of higher tax revenues (about 2⅓ percent of GDP) and lower expenditures (about 2⅓ percent of GDP). The fiscal consolidation contains high-quality measures: on the revenue side, it is based on raising tax rates and anti-evasion measures; on the spending side, it involves rightsizing public employment and reducing the role of the state on key economic sectors such as energy and telecommunications. To lock-in the gains from fiscal consolidation and make more institutional fiscal discipline/transparency and enhance credibility, the authorities have decided to introduce a fiscal responsibility law.

6. Public sector financing. Financing of the public sector is assumed to come mainly from external sources with a gradual increase in domestic sources. It is, however, assumed that the financing mix changes toward less concessional sources compared to recent years (Table A2), including within multilateral institutions. Due to lower financing needs relative to the last DSA, no

[5] The primary balance is measured as in the fiscal accounts deducting from the overall balance not only the interest payments on government liabilities but also the interest revenues on government assets.

placement of external bonds is included in the current DSA. Other financing needs are met through official, mostly multilateral, financing. Thanks to strong program implementation and Honduras' recent upgrade in sovereign ratings, relative to the last Eurobond issuance the authorities have recently received bond issuance offers at much more favorable conditions, including longer maturities. For now they are, however, opting to postpone issuing new external bonds but they could consider a debt management operation that significantly reduce the liquidity risk in 2020.[6] Against this background, for the DSA simulations in 2020 we assumed the same financing mix as in previous years.



7. External sector. The external current account deficit is projected to decline from 7.4 percent in 2014 to 5.8 percent by 2017, and stabilize at about 5.4 percent of GDP over the longer term. The previous DSA projected a weaker external position—with the current account deficit stabilizing at about 5.2 percent of GDP over the medium term—reflecting mainly higher oil prices in the short to medium term. The improvement in the current account during the program period reflects more favorable external conditions (including lower oil prices and continued recovery of worker remittances), and a stronger macroeconomic policy mix (fiscal consolidation after strong implementation of the program). As in the previous DSA, the current account deficit continues to be financed through foreign direct investment and, to a lesser extent, public sector borrowing, with very small private sector flows. This allows international reserves to increase to about 4.8 months of imports by 2017 and settle at close to 6 months of imports over the longer term.

[6] This potential debt management operation would not impact on the external DSA.

Table: Selected Macroeconomic Indicators, Current vs. Previous DSA

	2014	2015	2016	Long term 1/
Real GDP growth (percent)				
Current DSA	3.1	3.5	3.5	4.0
Previous DSA	3.0	3.0	3.3	3.7
Primary Balance (% GDP)				
Current DSA	-3.8	-1.2	-0.4	0.2
Previous DSA	-5.5	-2.9	-2.1	-0.4
Current account balance (% GDP)				
Current DSA	-7.4	-6.0	-5.8	-5.4
Previous DSA	-7.8	-7.1	-6.8	-5.2
Net FDI (% GDP)				
Current DSA	5.7	5.7	5.8	5.7
Previous DSA	5.0	5.0	4.9	4.8

1/ Defined as the last 15 years of the projection period. For the current DSA, the long term covers the period 2021-35, whereas for the previous DSA it covered 2020-34.

C. Public and External Debt Sustainability Analysis

8. Public debt ratios are expected to peak in 2018 and then start to decline subsequently (Table A1)**.** Public debt is projected to peak at about 51 percent of GDP in 2018 (up from 46.4 percent of GDP in 2014) and start falling slowly as fiscal consolidation proceeds and interest payments decline reaching 34 percent of GDP by 2027. This represents an improvement from the previous DSA, which projected a public debt-to-GDP ratio of 42 percent by that year. In present value terms, the public-debt ratio is expected to peak at 45 percent of GDP in 2018 and fall to about 20 percent of GDP in the late 2020s. The public debt dynamics remains somewhat vulnerable to both policy-related and exogenous shocks, especially to those related to economic growth and fiscal policy (Table A3).

9. The exposure to contingent liabilities seems to be limited. Several measures recently taken by the authorities limit the exposure of public debt to contingent liabilities. For instance, by doubling the contribution rate for the social security institute, the authorities' have improved its actuarial position; they amended the PPP framework law to eliminate the provision of government guarantees for PPP operations; and by upgrading to international standards the PPPs accounting (see section A), they have made transparent the potential fiscal impact of PPP's operations. There could, however, be a contingent obligation for ENEE if the interpretation of financial penalties for differing payments to energy providers, currently under litigation materialized. That said, this liability would be small. During 2009–14, the nominal amount under dispute is about 0.6 percent of GDP,

plus indemnification costs that are to be decided by the courts. As part of the implementation of a medium-term fiscal framework, the authorities are developing a comprehensive framework to analyze and determine potential fiscal risks.[7] It is unclear at the time this DSA was prepared if contingent obligations could arise from measures related to situation of the *Grupo Continental*.

10. The evolution of external debt ratios is projected to follow a similar pattern (Table A2). External debt ratios are projected to peak at 39.1 percent of GDP in 2017, from about 37.5 percent of GDP in 2014. The share of external debt in total public debt rises slowly to about two-thirds by the mid-2020s. Afterwards, it increases as total debt decreases substantially due to fiscal consolidation. Private external debt is less than 10 percent of GDP and is expected to shrink with the ongoing de-dollarization of the economy. The ratios for the PV of public debt and public debt service remain below their indicative thresholds in the baseline scenario throughout the projection period. (Table A4). No alternative scenario leads to breaching the thresholds for the PV of debt, as it was the case in the previous DSA. The threshold for the ratio of debt service to revenue is exceeded in only one year (2020) under the most extreme shock,[8] an improvement with respect to the previous DSA. The spikes reflect the amortization of a global bond with a balloon payment, which the authorities are planning to refinance to take advantage of improved financing conditions given Honduras' recent upgrade in sovereign ratings, as discussed above. The ratio of debt service to exports and remittances remain well below its threshold under all scenarios. External public debt rises initially but remains below the indicative threshold and starts to decline before 2020, earlier than in the previous DSA due to stronger fiscal consolidation and more favorable external conditions. The threshold for the ratio of debt service to revenue is exceeded once only (in 2020, owing to the global bond repayment, as mentioned above).[9]

11. A customized scenario with several negative shocks shows that external public debt indicators remain below their indicative thresholds. The scenario aims at capturing key risks and vulnerabilities in Honduras. It includes negative shocks to GDP growth, which could arise from weaker external conditions; tighter external financing conditions that could be associated with negative developments in international financial markets such as Fed tapering and EM turbulence; a weaker fiscal position from possible slippages in the implementation of adjustment measures. In this scenario, external public debt rises initially but remains below the indicative threshold and starts to decline around the mid-2020s. The ratio of debt service to exports and remittances, and to revenue also remain below its indicative threshold, while a similar customized scenario in the previous DSA presented breaches of the debt service to revenue ratio.

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[7] Last May, in the published Medium Term Fiscal Framework the authorities included a section on fiscal risks.

[8] This standardized stress test that simulates a 30 percent depreciation of the currency may overstate risks in a country like Honduras with a longstanding stable US dollar exchange rate whose external debt is denominated primarily in US dollars. Under this same scenario, the probability of external debt distress breaches temporarily the corresponding thresholds in the case of the PV of public debt and in the case of the ratio of debt service to revenue.

[9] In the last three years, Remittances were around 18 percent of GDP—still slightly below pre-global crisis levels but expected to continue to recover over the next few years—and close to 40 percent of exports of goods and services. Figure A2 shows that the same results hold after the inclusion of remittances.

Table: Baseline vs. Customized Scenario 1/

	2015	2016	2017	Long term 2/
Real GDP growth (percent)				
Baseline	3.5	3.5	3.7	4.0
Customized	3.1	2.6	2.5	3.5
Average interest rate on new external debt				
Baseline	3.8	3.8	3.8	3.8
Customized	4.9	4.9	4.9	4.9
Fiscal balance (% GDP)				
Baseline	-2.5	-2.0	-1.9	-1.0
Customized	-4.5	-3.7	-3.3	-1.3

1/ In the customized scenario, (i) real GDP growth is 3.1 perent in 2015, 2.6 percent in 2016 and 2.5 in 2017, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 106 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.

2/ Defined as the last 15 years of the projection period. For the current DSA, 2021-35.

D. Conclusion

12. Despite improvements on public and external debt indicators and strong program implementation we have decided to maintain Honduras risk of debt distress as moderate. The improvement in debt indicators reflect the stronger macroeconomic policies underpinning the IMF-supported program, which has been translated into substantially lower fiscal and current account deficits. This is also aided by stronger economic growth compared to the previous DSA. External public debt indicators are expected to stay below their indicative thresholds (except for the ratio of debt service to revenue under the most extreme shock), provided that the program's fiscal consolidation and growth targets are achieved. However, the macroeconomic consequences of the designation of some companies within the GC are difficult to fully assess at this stage. In this light, we have maintained our previous assessment of Honduras risk of debt distress as moderate. At the same time, should the program's fiscal consolidation and growth targets are achieved an improved rating of low risk of debt distress should be considered in a future DSA.

Table A1. Honduras: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012-2035

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	35.2	45.7	46.4			49.1	50.3	50.9	51.2	50.9	50.0		38.7	19.7	
of which: foreign-currency denominated	20.2	28.5	29.2			31.0	31.5	32.4	32.5	32.7	32.6		28.7	16.6	
Change in public sector debt	3.0	10.5	0.7			2.7	1.2	0.6	0.3	-0.3	-0.8		-2.4	-1.6	
Identified debt-creating flows	4.0	8.3	3.3			2.0	1.7	1.2	0.6	0.0	0.0		-0.9	-0.1	
Primary deficit 2/	4.3	7.2	3.8	3.7	1.6	1.2	0.4	0.3	-0.1	-0.5	-0.3	0.2	-0.4	0.2	-0.2
Revenue and grants	21.1	21.4	22.8			24.8	25.0	25.2	25.2	25.4	25.5		25.5	26.3	
of which: grants	1.0	0.8	0.8			0.8	0.7	0.6	0.6	0.5	0.4		0.4	0.4	
Primary (noninterest) expenditure	25.5	28.6	26.6			26.0	25.3	25.5	25.1	24.8	25.2		25.1	26.5	
Automatic debt dynamics	-0.3	1.1	-0.5			0.8	1.3	0.9	0.7	0.5	0.3		-0.5	-0.3	
Contribution from interest rate/growth differential	-0.8	0.4	-0.6			0.4	0.5	0.1	-0.2	-0.4	-0.6		-0.6	0.0	
of which: contribution from average real interest rate	0.4	1.4	0.8			2.0	2.2	1.9	1.6	1.6	1.4		1.0	0.8	
of which: contribution from real GDP growth	-1.3	-1.0	-1.4			-1.6	-1.7	-1.8	-1.9	-2.0	-2.0		-1.6	-0.8	
Contribution from real exchange rate depreciation	0.5	0.7	0.1			0.4	0.8	0.9	0.9	0.9	0.9		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.0	2.2	-2.6			0.8	-0.5	-0.6	-0.3	-0.3	-0.8		-1.5	-1.5	
Other Sustainability Indicators															
PV of public sector debt	...	...	41.0			43.3	44.5	45.1	45.4	44.8	43.6		32.5	15.7	
of which: foreign-currency denominated	...	...	23.9			25.1	25.8	26.6	26.7	26.6	26.2		22.5	12.6	
of which: external	...	...	23.9			25.1	25.8	26.6	26.7	26.6	26.2		22.5	12.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	4.6	8.0	4.8			6.3	4.8	5.0	7.2	7.8	8.6		4.7	1.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	180.2			174.6	178.4	178.8	180.3	176.8	171.0		127.2	59.6	
PV of public sector debt-to-revenue ratio (in percent)	...	...	186.6			180.2	183.8	183.5	184.5	180.3	174.0		129.4	60.6	
of which: external 4/	...	...	108.5			104.6	106.3	108.3	108.6	107.1	104.3		89.7	48.7	
Debt service-to-revenue and grants ratio (in percent) 5/	7.2	9.9	10.4			25.3	22.3	23.2	33.0	36.6	38.6		24.6	9.5	
Debt service-to-revenue ratio (in percent) 6/	7.5	10.3	10.8			26.1	22.9	23.7	33.7	37.2	39.2		25.0	9.6	
Primary deficit that stabilizes the debt-to-GDP ratio	1.3	-3.3	3.1			-1.6	-0.8	-0.3	-0.4	-0.2	0.5		2.0	1.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	4.1	2.8	3.1	3.8	2.6	3.5	3.5	3.7	3.8	4.0	4.0	3.8	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	1.8	2.0	3.3	1.8	0.6	3.5	3.5	3.5	3.5	3.4	3.5	3.5	3.0	2.9	3.0
Average real interest rate on domestic debt (in percent)	3.3	9.0	1.8	0.9	3.6	7.6	8.6	7.5	6.8	6.6	5.4	7.1	6.5	8.6	6.5
Real exchange rate depreciation (in percent, + indicates depreciation)	3.0	3.4	0.2	-2.2	3.4	1.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.6	1.4	5.5	5.8	2.1	3.6	3.6	3.9	4.3	4.3	4.3	4.0	4.2	1.6	4.0
Growth of real primary spending (deflated by GDP deflator, in percent)	6.7	15.6	-4.2	2.0	5.8	0.9	1.1	4.2	2.2	3.0	5.5	2.8	4.7	7.1	4.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	22.6	10.8	14.9	16.7	23.9	17.0	17.7	17.0	17.0	...

Sources: Country authorities; and staff estimates and projections.
1/ Gross debt of the Combined Public Sector
2/ As defined in the fiscal accounts
3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
4/ Revenues excluding grants.
5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A2. Honduras: External Debt Sustainability Framework, Baseline Scenario, 2012-2035 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**26.7**	**36.3**	**37.5**			**38.7**	**38.7**	**39.1**	**38.6**	**38.2**	**38.1**		**34.0**	**21.4**	
of which: public and publicly guaranteed (PPG)	20.2	28.5	29.2			31.0	31.5	32.4	32.5	32.7	32.6		28.7	16.6	
Change in external debt	2.8	9.6	1.2			1.2	0.0	0.3	-0.4	-0.4	-0.1		-1.3	-1.2	
Identified net debt-creating flows	2.8	4.2	-0.2			-0.9	-1.3	-1.3	-1.5	-1.6	-1.6		-1.7	-1.7	
Non-interest current account deficit	**8.0**	**8.8**	**6.4**	**6.6**	**3.8**	**4.7**	**4.5**	**4.5**	**4.4**	**4.4**	**4.4**		**4.3**	**4.3**	4.4
Deficit in balance of goods and services	19.5	20.6	18.9			18.2	17.9	18.1	18.4	18.3	18.3		18.3	17.3	
Exports	50.9	47.7	46.9			46.2	46.7	47.9	49.3	50.5	51.7		53.4	50.6	
Imports	70.3	68.3	65.9			64.4	64.7	66.1	67.6	68.8	70.0		71.8	67.9	
Net current transfers (negative = inflow)	-17.8	-18.4	-18.3	-19.4	1.8	-19.3	-19.5	-19.7	-20.0	-20.0	-20.0		-20.0	-20.3	-20.0
of which: official	-1.2	-0.8	-0.9			-0.9	-0.9	-0.9	-0.9	-0.9	-0.9		-0.9	-0.9	
Other current account flows (negative = net inflow)	6.3	6.6	5.8			5.8	6.1	6.1	6.1	6.1	6.1		6.0	7.3	
Net FDI (negative = inflow)	**-4.6**	**-5.4**	**-5.7**	**-5.8**	**1.2**	**-5.7**	**-5.8**	**-5.8**	**-5.8**	**-5.8**	**-5.8**		**-5.7**	**-5.8**	-5.7
Endogenous debt dynamics 2/	**-0.6**	**0.7**	**-0.9**			**0.0**	**0.0**	**-0.1**	**-0.1**	**-0.2**	**-0.2**		**-0.3**	**-0.2**	
Contribution from nominal interest rate	0.5	0.7	1.0			1.3	1.3	1.3	1.3	1.3	1.3		1.1	0.7	
Contribution from real GDP growth	-0.9	-0.7	-1.1			-1.3	-1.3	-1.4	-1.4	-1.5	-1.5		-1.3	-0.9	
Contribution from price and exchange rate changes	-0.2	0.8	-0.8			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**0.0**	**5.4**	**1.4**			**2.1**	**1.3**	**1.6**	**1.0**	**1.2**	**1.5**		**0.4**	**0.5**	
of which: exceptional financing	0.0	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	32.1			32.8	32.9	33.2	32.8	32.2	31.7		27.8	17.4	
In percent of exports	...	...	68.5			71.1	70.5	69.3	66.5	63.8	61.3		52.0	34.4	
PV of PPG external debt	**...**	**...**	**23.9**			**25.1**	**25.8**	**26.6**	**26.7**	**26.6**	**26.2**		**22.5**	**12.6**	
In percent of exports	**...**	**...**	**50.8**			**54.4**	**55.1**	**55.5**	**54.2**	**52.8**	**50.6**		**42.1**	**25.0**	
In percent of government revenues	**...**	**...**	**101.0**			**98.2**	**100.2**	**102.1**	**102.4**	**101.1**	**98.5**		**84.7**	**46.0**	
Debt service-to-exports ratio (in percent)	**15.8**	**16.9**	**15.3**			**17.8**	**16.9**	**15.8**	**14.8**	**13.8**	**17.0**		**13.4**	**10.3**	
PPG debt service-to-exports ratio (in percent)	**1.4**	**1.6**	**2.9**			**3.7**	**4.0**	**4.1**	**4.4**	**4.5**	**8.7**		**5.8**	**3.0**	
PPG debt service-to-revenue ratio (in percent)	**3.4**	**3.5**	**5.8**			**6.6**	**7.2**	**7.6**	**8.2**	**8.6**	**16.9**		**11.6**	**5.5**	
Total gross financing need (Billions of U.S. dollars)	2.4	2.4	1.8			1.8	1.7	1.7	1.6	1.5	2.0		2.0	2.4	
Non-interest current account deficit that stabilizes debt ratio	5.2	-0.8	5.2			3.5	4.5	4.2	4.9	4.8	4.5		5.6	5.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	4.1	2.8	3.1	3.8	2.6	3.5	3.5	3.7	3.8	4.0	4.0	3.8	4.0	4.0	4.0
GDP deflator in US dollar terms (change in percent)	0.7	-2.8	2.3	4.4	3.5	-0.9	-0.9	-1.0	-0.7	-0.7	-0.6	-0.8	1.6	1.6	1.3
Effective interest rate (percent) 5/	2.3	2.7	2.9	2.4	0.4	3.6	3.5	3.5	3.5	3.5	3.5	3.5	3.1	3.1	3.2
Growth of exports of G&S (US dollar terms, in percent)	4.6	-6.3	3.9	6.6	13.0	0.9	3.8	5.2	6.0	5.8	5.8	4.6	5.7	0.0	5.3
Growth of imports of G&S (US dollar terms, in percent)	3.5	-3.0	1.7	8.0	15.7	0.2	3.0	4.9	5.5	5.0	5.1	4.0	5.7	0.0	5.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	22.6	10.8	14.9	16.7	23.9	17.0	17.7	17.0	17.0	17.0
Government revenues (excluding grants, in percent of GDP)	21.5	22.2	23.6			25.6	25.7	26.0	26.1	26.3	26.6		26.6	27.4	26.6
Aid flows (in Billions of US dollars) 7/	0.4	0.4	0.3			0.3	0.2	0.3	0.2	0.1	0.2		0.2	0.3	
of which: Grants	0.2	0.1	0.2			0.2	0.1	0.1	0.1	0.1	0.1		0.1	0.2	
of which: Concessional loans	0.2	0.2	0.2			0.1	0.1	0.1	0.0	0.0	0.1		0.1	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.5	0.9	1.0	0.9	1.1	1.2		0.9	0.6	0.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			37.9	34.5	31.9	34.1	37.4	24.5		29.2	42.7	36.6
Memorandum items:															
Nominal GDP (Billions of US dollars)	18.5	18.5	19.5			20.0	20.5	21.1	21.7	22.4	23.2		29.3	51.0	
Nominal dollar GDP growth	4.9	-0.1	5.5			2.5	2.6	2.6	3.1	3.3	3.3	2.9	5.7	5.7	5.4
PV of PPG external debt (in Billions of US dollars)			4.5			4.9	5.2	5.5	5.7	5.8	5.9		6.5	6.4	
(PVt-PVt-1)/GDPt-1 (in percent)						2.0	1.2	1.5	0.9	0.8	0.4	1.1	0.1	-0.1	0.1
Gross workers' remittances (Billions of US dollars)	2.9	3.1	3.4			3.7	3.8	4.0	4.1	4.3	4.4		5.6	9.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	20.4			21.2	21.7	22.4	22.4	22.4	22.0		18.9	10.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	37.2			39.0	39.5	39.9	39.1	38.4	37.0		31.1	18.4	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.1			2.6	2.8	3.0	3.1	3.3	6.4		4.3	2.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

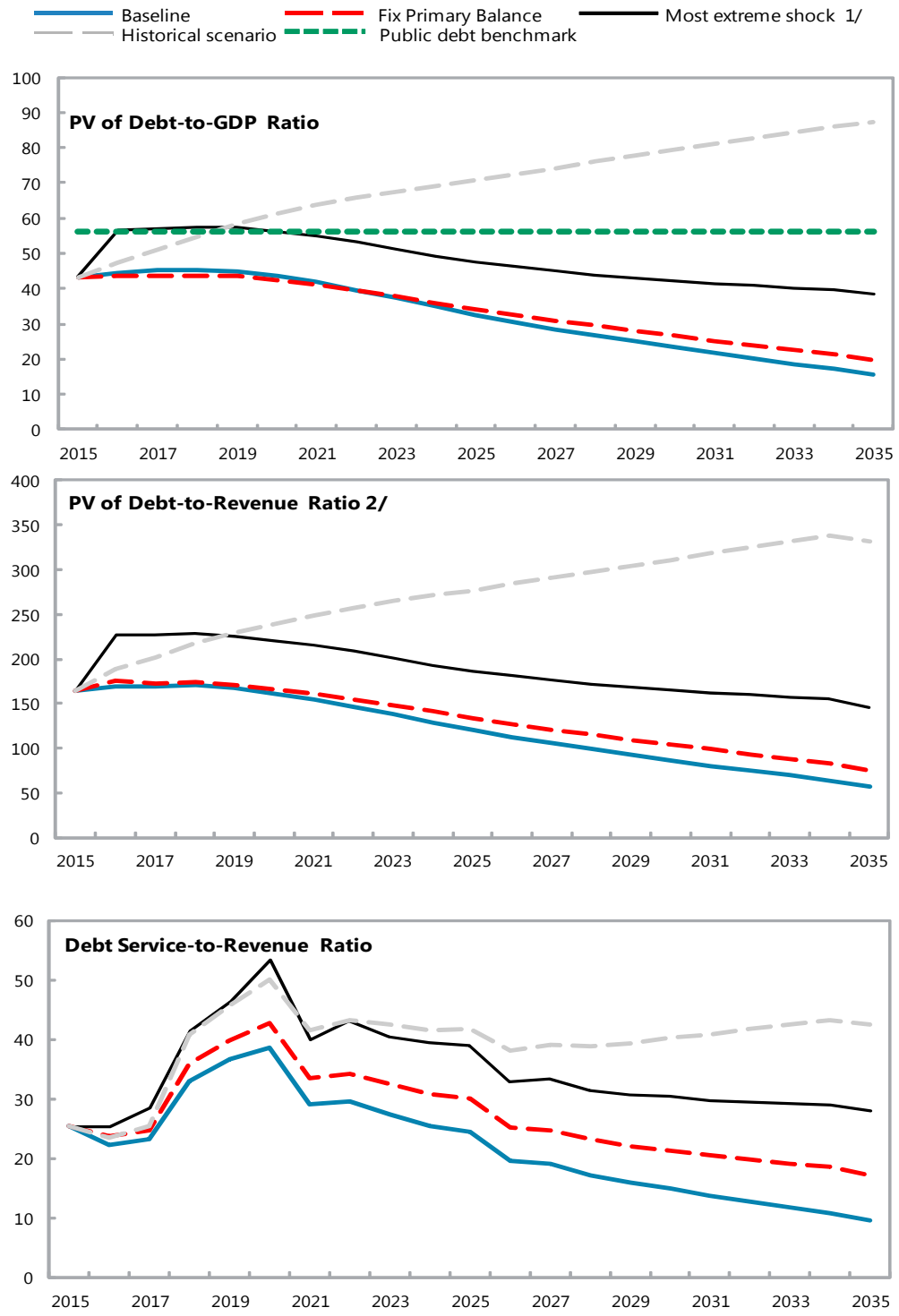
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A3. Honduras: Sensitivity Analysis for Key Indicators of Public Debt 2015-2035

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	43	45	45	45	45	44	32	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	43	47	51	55	58	61	71	87
A2. Primary balance is unchanged from 2015	43	44	44	44	43	42	34	20
A3. Permanently lower GDP growth 1/	43	45	46	47	47	47	44	53
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	43	46	49	50	51	51	46	44
B2. Primary balance is at historical average minus one standard deviations in 2016-201	43	49	54	54	54	53	44	32
B3. Combination of B1-B2 using one half standard deviation shocks	43	49	54	54	55	54	47	39
B4. One-time 30 percent real depreciation in 2016	43	56	57	57	57	56	47	38
B5. 10 percent of GDP increase in other debt-creating flows in 2016	43	53	54	54	54	53	44	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	164	168	169	170	167	162	120	56
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	164	189	202	217	230	240	277	332
A2. Primary balance is unchanged from 2015	164	175	173	174	171	166	134	74
A3. Permanently lower GDP growth 1/	164	179	182	186	186	185	172	201
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	164	184	192	199	200	199	181	165
B2. Primary balance is at historical average minus one standard deviations in 2016-201	164	196	214	216	213	209	172	120
B3. Combination of B1-B2 using one half standard deviation shocks	164	194	212	216	215	212	184	146
B4. One-time 30 percent real depreciation in 2016	164	226	226	229	226	220	186	146
B5. 10 percent of GDP increase in other debt-creating flows in 2016	164	214	215	216	214	209	172	120
Debt Service-to-Revenue Ratio 2/								
Baseline	25	22	23	33	37	39	25	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	24	25	41	46	50	42	42
A2. Primary balance is unchanged from 2015	25	24	25	36	40	43	30	17
A3. Permanently lower GDP growth 1/	25	24	25	38	42	45	34	30
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	25	24	26	39	44	47	35	26
B2. Primary balance is at historical average minus one standard deviations in 2016-201	25	24	26	43	47	46	32	22
B3. Combination of B1-B2 using one half standard deviation shocks	25	24	26	43	47	47	34	25
B4. One-time 30 percent real depreciation in 2016	25	25	29	41	46	53	39	28
B5. 10 percent of GDP increase in other debt-creating flows in 2016	25	24	27	47	43	47	32	22

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure A1. Honduras: Indicators of Public Debt Under Alternative Scenarios, 2015-2035 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Table A4. Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015-2035

(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to-GDP+remittances ratio								
Baseline	21	22	22	22	22	22	**19**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	21	22	23	24	24	25	**26**	24
A2. New public sector loans on less favorable terms in 2015-2035 2	21	22	23	23	24	24	**24**	18
A3. Customized 7/	21	22	24	25	26	27	27	20
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	21	22	23	23	23	22	**19**	11
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	21	24	31	31	31	31	**28**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	21	21	21	21	21	21	**18**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	21	25	29	28	28	28	**24**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	21	23	26	26	26	26	**23**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	21	28	29	29	29	29	**25**	14
PV of debt-to-exports+remittances ratio								
Baseline	39	39	40	39	38	37	**31**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	39	41	42	42	43	43	**45**	44
A2. New public sector loans on less favorable terms in 2015-2035 2	39	39	41	41	41	41	**40**	32
A3. Customized 7/	45	48	50	51	52	52	51	40
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	39	39	39	38	37	36	**30**	18
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	39	47	64	63	62	60	**53**	28
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	39	39	39	38	37	36	**30**	18
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	39	46	52	49	48	46	**40**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	39	44	52	50	49	48	**41**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	39	39	39	38	37	36	**30**	18
PV of debt-to-revenue ratio								
Baseline	98	100	102	102	101	98	**85**	46
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	98	102	105	106	107	107	**112**	95
A2. New public sector loans on less favorable terms in 2015-2035 2	98	100	104	106	107	109	**108**	79
A3. Customized 7/	96	86	93	97	100	102	103	73
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	98	100	104	104	102	100	**86**	47
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	98	112	142	143	142	139	**125**	60
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	98	96	96	97	95	93	**80**	43
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	98	111	127	127	126	124	**110**	54
B5. Combination of B1-B4 using one-half standard deviation shocks	98	104	117	117	116	113	**101**	50
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	98	140	143	143	141	137	**119**	64

Table A4. Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015-2035 (continued)

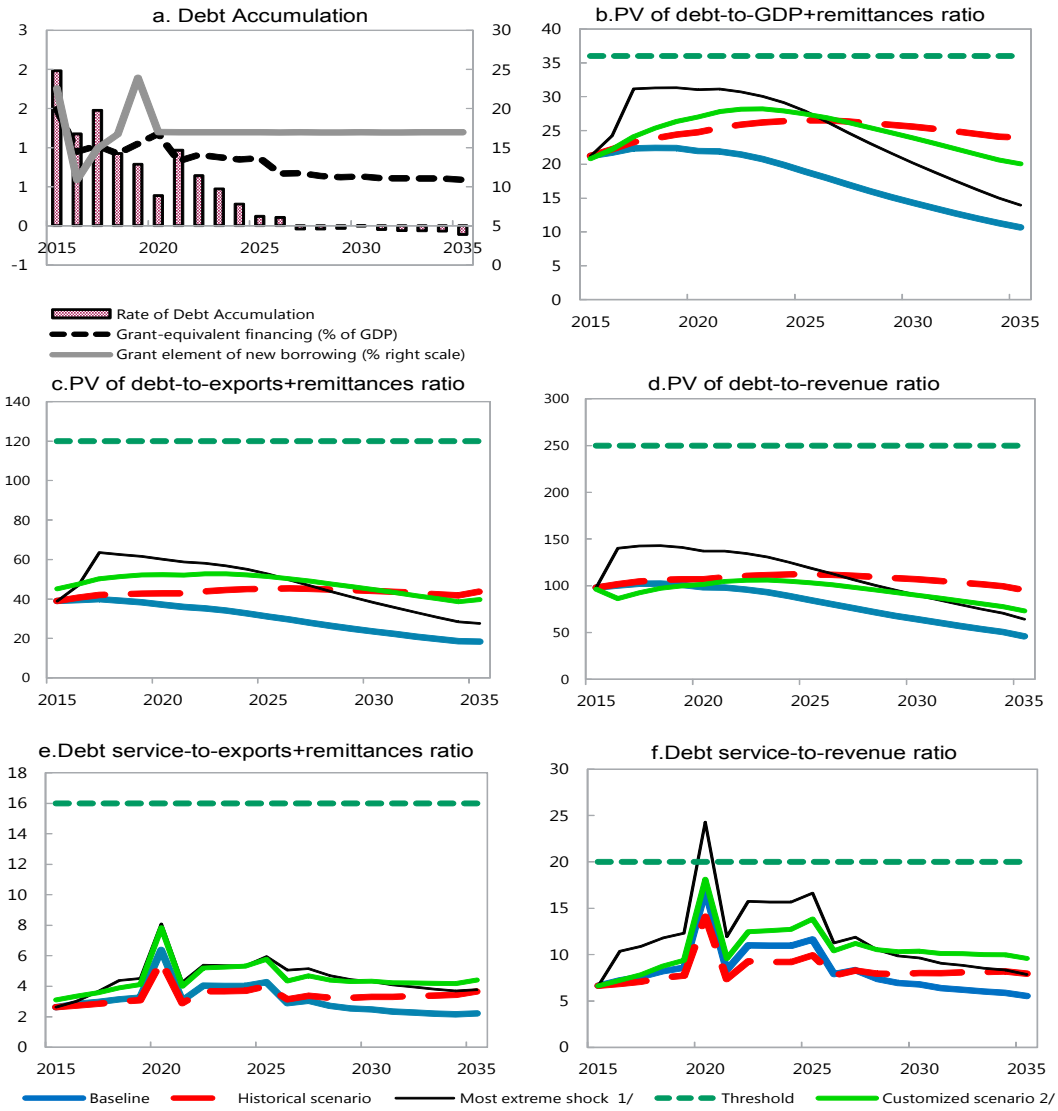
(In percent)

Debt service-to-exports+remittances ratio								
Baseline	3	3	3	3	3	6	**4**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	3	3	3	3	3	6	**4**	4
A2. New public sector loans on less favorable terms in 2015-2035 2	3	3	3	3	3	6	**4**	3
A3. Customized 7/	3	3	4	4	4	8	6	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	3	3	3	3	3	6	**4**	2
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	3	3	4	4	5	8	**6**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	3	3	3	3	3	6	**4**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	3	3	3	4	4	7	**5**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	3	4	4	7	**5**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	3	3	3	3	3	6	**4**	2
Debt service-to-revenue ratio								
Baseline	7	7	8	8	9	17	**12**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	7	7	7	8	8	14	**10**	8
A2. New public sector loans on less favorable terms in 2015-2035 2	7	7	7	8	8	16	**11**	8
A3. Customized 7/	7	7	8	9	9	18	14	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	7	7	8	9	9	18	**12**	6
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	7	7	8	10	10	19	**14**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	7	7	7	8	8	16	**11**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	7	7	8	9	10	18	**13**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	7	7	8	9	9	17	**12**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	7	10	11	12	12	24	**17**	8
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	12	12	12	12	12	12	**12**	12

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.
7/ In the customized scenario, (i) real GDP growth is 3.1 perent in 2015, 2.6 percent in 2016 and 2.5 in 2017, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 106 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.

Figure A2. Honduras: Indicators of Public and Publicly Guaranteed External Debt Including Remittances, 2015-2035 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

2/ In the customized scenario, (i) real GDP growth is 3.1 perent in 2015, 2.6 percent in 2016 and 2.5 in 2017, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 106 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.